

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-7960
Project Name	Import Data using Transform Maps (Spreadsheet) in ServiceNow
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Manual Spreadsheet Uploads	Requires human intervention to upload CSV/Excel files for every roster update.	Medium
2	Basic Error Logging	Failed transforms require manual inspection of Transform History logs without instant automated email alerts.	Low
3	Single-source File Processing	Processes one spreadsheet file at a time rather than automated multi-source batch ingestion.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handled by standalone admin users accessing the staging modules.	Scale to support multi-department HR teams concurrently managing distinct regional employee datasets.
2	Data Storage	Handles standard batch sizes (up to 1,000+ rows) in custom staging tables.	Implement automated table archiving and data retention policies to optimize storage performance.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	Performance	Executes synchronous transform mapping for current batch workloads.	Implement asynchronous scheduled import sets and background worker queues for bulk data processing.
4	Security	Role-Based Access Control (RBAC) restricted to instance administrators.	Integrate advanced OAuth/SSO token authentication and field-level encryption for sensitive PII data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated Scheduled Imports (SFTP / REST API Integration)	Q4 2026	Eliminates manual spreadsheet uploads by automatically fetching employee data feeds from external HR systems.
Phase 3	Advanced Error Notification & Notification Workflows	Q1 2027	Instantly notifies administrators via email/Slack when transform errors or mandatory field mismatches occur.
Phase 4	Predictive Analytics & AI-driven Data Validation	Q2 2027	Automatically detects anomalies, formatting inconsistencies, and suggests field mapping corrections prior to transformation.